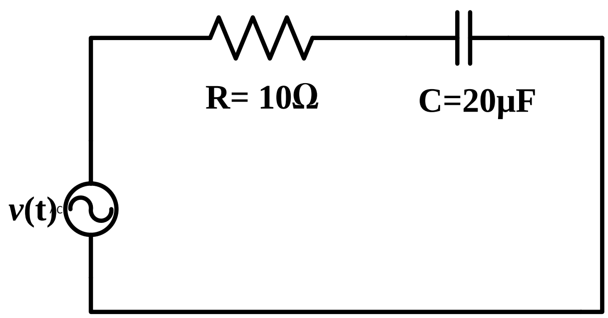
ABES Engineering College, Ghaziabad

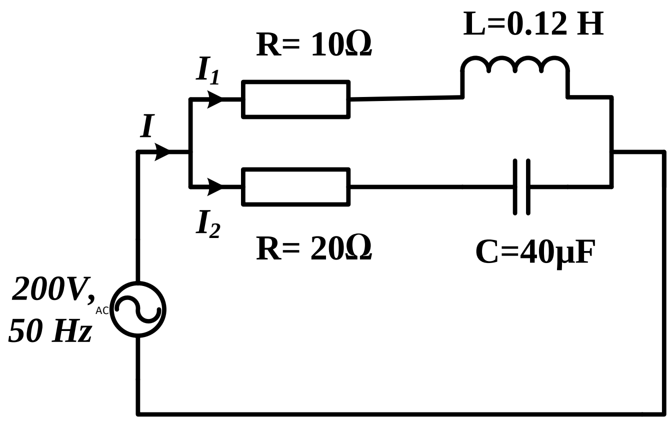
Department of Applied sciences & Humanities

Session: 2025-26 **Semester:** I **Section:**

Course Code: 25EL101 **Course Name:** Basics of Electrical Engineering

Assignment Sheet-2 (Unit-2)

Q. No.	KL	CO	Question	Ans.
Q-1	**K3**	**CO1**	In the given RC circuit, the current $i(t) = 2cos5000t$. Find the (a) equivalent impedance, (b) rms current, and rms voltage, (c) expression for instantaneous voltage, (d) power factor, (e) active, power delivered by the source. 	**(a) 14.14∠-45°Ω (b) 1.41A,20 V (c) 28.28 cos(5000t-45°) (d) 0.7 leading (e) 20 W**
Q-2	**K3**	**CO1**	For an ac series circuit, if $v(t) = 160\sin(\omega t + 10^{o})$ and $i(t) = 5\sin(\omega t - 20^{o})$. Find the (a) power factor, (b) active power, (c) reactive power, and (d) apparent power absorbed in the circuit. (e) Draw the phasor diagram and power triangle for the circuit.	**(a) 0.86 lag (b) 346.4 W (c) 200 VAR (d) 400 VA**
Q-3	**K3**	**CO1**	A parallel RL circuit, $R = 50\Omega$, $X_L = 40\Omega$ is connected across the 100 V ac voltage source. Calculate (a) current from the source, (b) real power, (c) reactive power, and (d) apparent power.	**(a) 3.2 A, (b) 200 W, (c) 249.7 VAR, (d) 320 VA**

Q-4	**K3**	**CO1**	Two branches are connected in parallel to 200 V, 50 Hz source as shown in the figure. Find (a) total impedance, (b) total admittance, (c) current in each branch, (d) the source current, (e) power factor. 	**(a) 62.1∠53.6°Ω (b) 0.016∠-53.6° Ω^{-1} (c) I_1=5.13∠-75.14°A, I_2=2.44∠75.89°A (d) 3.22∠-53.6° (e) 0.59 lagging**
Q-5	**K3**	**CO1**	A voltage $v(t) = 100\sin 314t$ is applied to a series circuit consisting of 10 Ω resistance, 0.0318 H inductance, and a capacitance of 63.6 μF. Calculate (a) total impedance of the circuit, (b) rms value of voltage and current, (c) instantaneous current expression, (d) power factor, (e) active, reactive and apparent power.	**(a) 41.32∠-76 °Ω (b) 70.71V, 1.71A, (c) 2.42 $\sin(314t + 76°)$ (d) 0.24 leading (e) 29.25 W, 117.32 VAR, 120.91 VA**
Q-6	**K3**	**CO1**	A coil having an inductance of 50 mH and resistance 10 Ω is connected in series with a 25 μF capacitor across a 200 V ac supply. Calculate (a) resonance frequency of the circuit (b) current flowing at resonance and (c) quality factor, (d) upper and lower corner frequencies, (e) bandwidth.	**(a) 142.35Hz (b) 20 A (c) 4.47 (d) 158.26Hz, 126.43 Hz (e) 31.83 Hz**
Q-7	**K3**	**CO1**	A 400V, 3-phase supply is connected to a balanced network of three impedances each consisting of a 20 Ω resistance and 15 Ω inductive reactance. Determine (a) impedance in each branch, (b) phase and line current, (c) total active, reactive and apparent power when the three impedances are connected in star.	**(a) 25∠36.87°Ω (b) 9.24A, 9.24A (c)5.12 kW, 3.84 kVAR, 6.4 kVA**

Q-8	**K3**	**CO1**	A 400V, 3-phase supply is connected to a balanced network of three impedances each consisting of a 20 Ω resistance and 15 Ω inductive reactance. Determine (a) impedance in each branch, (b) phase and line current, (c) total active, reactive and apparent power when the three impedances are connected in delta.	**(a) $25\angle 36.87^\circ$ Ω (b) 16A, 27.71A (c) 15.35 kW, 11.5 kVAR, 19.2 kVA**
Q-9	**K3**	**CO1**	A resistance R, capacitance C and inductance L of value 0.01 H are connected in series. When a voltage $v(t) = 400\cos(3000t - 10^\circ)$ V is applied to the series combination. the current flowing is given by $i(t) = 10\sqrt{2}\cos(3000\mathrm{t} - 55^\circ)$ A. Find the value of R and C.	**R = 20 Ω, C = 33.33 μF**
Q-10	**K3**	**CO1**	A series combination of R and C is in parallel with 20 Ω resistance across 50 Hz supply. If the total current is 7A, Current through 20 Ω resistor is 5A and current in R-C branch is 3A. Find values of R & C	**R =16.67 Ω C =110.322 μF**